TELECOM LAB | LAST-MINUTE SHEET

Experiments 1, 6, 7, 12 - formulas, expected outputs and viva triggers

EXPERIMENT 1 - PCM

Aim: Convert analog samples to binary PCM and reconstruct them.

A=4, fm=2 Hz, fs=200 Hz; 8 levels => 3 bits/sample.

fs >= 2 fmax; bitrate = fs x 3 = 600 bits/s; 2.2 -> 2.5 -> 110.

Show: original, sampled, quantized, PCM bits, decoded, LPF output.

EXPERIMENT 6 - DSSS

Aim: Spread each data bit with a faster PN-chip sequence.

Data 101; PN 0010111; bipolar 0=-1, 1=+1.

7 chips per bit => 7-fold spreading, PG = 8.45 dB.

Show: baseband spread, carrier, DSSS-BPSK; recovered data 101.

EXPERIMENT 7 - FHSS

Aim: Change frequency in a sequence to spread transmission.

7 hop frequencies; PN/hop control and RX sync are important.

FHSS vs DSSS: hop frequency vs multiply by PN chips.

Show: polar bits, BPSK, hopping carrier, FHSS waveform.

EXPERIMENT 12 - CONVOLUTIONAL

Aim: Generate redundant encoded bits using shift registers + XOR.

Input 1011, append 00; rate 1/2, memory 2, constraint length 3.

c1 = u XOR d1 XOR d2; c2 = u XOR d2.

Output pairs: 11 10 00 01 01 11; stream 111000010111.

Exam flow: AIM -> BLOCKS -> CODE -> OUTPUT -> RESULT -> VIVA

Run MATLAB .m if required; Python .py if permitted. Scripts + plots in accompanying ZIP.